

Signed sub-optimality bound for sample-path AIS

Extended Theorem 3

Lemma 1 (Policy evaluation). *Let $\hat{\pi} = (\hat{\pi}_1, \dots, \hat{\pi}_T)$, $\hat{\pi}_t : \mathcal{Z} \rightarrow \mathcal{A}$, be any AIS policy and $\pi_t = \pi_t \circ \sigma$, $\pi_t : \mathcal{H} \rightarrow \mathcal{A}$ its lifted version with $\pi_t(h_t) = \hat{\pi}_t(\sigma_t(h_t))$ for all $t \in \{1, \dots, T\}$. Define*

$$\alpha_t^{\hat{\pi}}(h_t) = V_t^{\pi}(h_t) - \hat{V}^{\hat{\pi}}(\sigma_t(h_t)), \quad (1)$$

$$\beta_t^{\hat{\pi}}(h_t, a_t) = Q_t^{\pi}(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t), \quad (2)$$

and

$$\Delta_t^{\hat{\pi}}(h_t, a_t) = \left[c_t(h_t, a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) \right] \quad (3)$$

$$- \left[\hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^{\sigma}(dz_{t+1} | h_t, a_t) \right]. \quad (4)$$

Then, for all $h_t \in \mathcal{H}_t$ and $a_t \in \mathcal{A}$, the following holds

$$\beta_t^{\hat{\pi}}(h_t, a_t) = \Delta_t^{\hat{\pi}}(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (5)$$

$$\alpha_t^{\hat{\pi}}(h_t) = \beta_t^{\hat{\pi}}(h_t, \hat{\pi}_t(\sigma_t(h_t))). \quad (6)$$

where $\alpha_{T+1}^{\hat{\pi}} \equiv 0$.

Proof. We first establish the recursion (5) and then the identity (6).

Step 1 (the $\beta^{\hat{\pi}}$ -recursion). Fix $t \in \{1, \dots, T\}$, $h_t \in \mathcal{H}_t$, and $a_t \in \mathcal{A}$. Adding and subtracting $\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t)$, and using $\hat{P}_t(\cdot | \sigma_t(h_t), a_t) = \hat{P}_t(\cdot | h_t, a_t)$, we get

$$\begin{aligned} \beta_t^{\hat{\pi}}(h_t, a_t) &= Q_t^{\pi}(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t) \\ &= [c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)] \\ &\quad + \int V_{t+1}^{\pi}(h_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t(dz_{t+1} | \sigma_t(h_t), a_t) \\ &= \underbrace{[c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)] + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^{\sigma}(dz_{t+1} | h_t, a_t)}_{\Delta_t^{\hat{\pi}}(h_t, a_t)} \\ &\quad + \int V_{t+1}^{\pi}(h_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t), \end{aligned}$$

where the first bracket is $\Delta_t^{\hat{\pi}}(h_t, a_t)$ by (3). For the remaining terms, we have

$$\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) = \int \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) P_{Y,t}(dy_{t+1} | h_t, a_t).$$

Thus the last term can be written as

$$\begin{aligned} & \int V_{t+1}^\pi(h_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) \\ & - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t) = \int \left[V_{t+1}^\pi(h_{t+1}) - \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) \right] P_{Y,t}(dy_{t+1} | h_t, a_t) \\ & = \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) \end{aligned}$$

so the two integrals against $P_{Y,t}$ combine into

$$\int \left[V_{t+1}^\pi(h_{t+1}) - \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) \right] P_{Y,t}(dy_{t+1} | h_t, a_t) = \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t),$$

This establishes (5). At $t = T$, we have $\alpha_{T+1}^{\hat{\pi}} \equiv 0$ and $\Delta_T^{\hat{\pi}}(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T)$, so (5) reads $\beta_T^{\hat{\pi}} = \Delta_T^{\hat{\pi}}$.

Step 2 (the $\alpha^{\hat{\pi}}$ -identity). Since $\hat{\pi}_t$ is deterministic and $\pi_t(h_t) = \hat{\pi}_t(\sigma_t(h_t))$, the value functions are the Q -functions evaluated at the prescribed actions:

$$V_t^\pi(h_t) = Q_t^\pi(h_t, \hat{\pi}_t(\sigma_t(h_t))), \quad \hat{V}_t^{\hat{\pi}}(\sigma_t(h_t)) = \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), \hat{\pi}_t(\sigma_t(h_t))).$$

Then, we can write

$$\alpha_t^{\hat{\pi}^*}(h_t) = V_t^\pi(h_t) - \hat{V}_t^{\hat{\pi}}(\sigma_t(h_t)) \tag{7}$$

$$= Q_t^\pi(h_t, \hat{\pi}_t(\sigma_t(h_t))) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), \hat{\pi}_t(\sigma_t(h_t))) \tag{8}$$

$$= \beta_t^{\hat{\pi}}(h_t, \hat{\pi}_t(\sigma_t(h_t))) \tag{9}$$

□

Lemma 2 (Optimality). *Define*

$$\alpha_t^*(h_t) = V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \tag{10}$$

$$\beta_t^*(h_t, a_t) = Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t), \tag{11}$$

and

$$\Delta_t(h_t, a_t) = \left[c_t(h_t, a_t) + \int \hat{V}_{t+1}^*(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t) \right] \tag{12}$$

$$- \left[\hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^*(z_{t+1}) \hat{P}_t^\sigma(dz_{t+1} | h_t, a_t) \right]. \tag{13}$$

Then, the following holds

$$\beta_t^*(h_t, a_t) = \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \tag{14}$$

$$\beta_t^*(h_t, \pi_t^*(h_t)) \leq \alpha_t^*(h_t) \leq \beta_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \tag{15}$$

where $\alpha_{T+1}^* \equiv 0$.

Proof. Step 1 (the β^ -recursion).* [Complete Step 1, similar to Lemma 1.1]

The derivation is identical to Step 1 in the proof of Lemma 1 with $(Q_t^\pi, \hat{Q}_t^\pi, V_{t+1}^\pi, \hat{V}_{t+1}^\pi)$ replaced by $(Q_t^*, \hat{Q}_t^*, V_{t+1}^*, \hat{V}_{t+1}^*)$, add and subtract $\int \hat{V}_{t+1}^*(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t)$, and identify the cost-and-kernel-mismatch bracket as $\Delta_t(h_t, a_t)$ from (12).

Step 2 (the α^ -inequalities).* The two inequalities in (15) follow from the two optimality properties, used one at a time.

Upper bound. By optimality of $\hat{\pi}_t^*$ on the AIS side, $\hat{V}_t^*(\sigma_t(h_t)) = \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t)))$, while on the true side $V_t^*(h_t) = \inf_{a \in \mathcal{A}} Q_t^*(h_t, a) \leq Q_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t)))$. Therefore,

$$\begin{aligned} \alpha_t^*(h_t) &= V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)) \\ &\leq Q_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) - \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t))) = \beta_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \end{aligned}$$

Lower bound. By optimality of π_t^* on the true side, $V_t^*(h_t) = Q_t^*(h_t, \pi_t^*(h_t))$, while on the AIS side $\hat{V}_t^*(\sigma_t(h_t)) = \inf_{a \in \mathcal{A}} \hat{Q}_t^*(\sigma_t(h_t), a) \leq \hat{Q}_t^*(\sigma_t(h_t), \pi_t^*(h_t))$. Therefore,

$$\begin{aligned} \alpha_t^*(h_t) &= V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)) \\ &\geq Q_t^*(h_t, \pi_t^*(h_t)) - \hat{Q}_t^*(\sigma_t(h_t), \pi_t^*(h_t)) = \beta_t^*(h_t, \pi_t^*(h_t)). \end{aligned}$$

Combining the two gives (15). $\square$

Theorem 3. Suppose that for each $t \in \{1, \dots, T\}$, there exist measurable functions $\alpha_t^{\hat{\pi}^*}: \mathcal{H}_t \rightarrow \mathbb{R}$, $\alpha_t^*: \mathcal{H}_t \rightarrow \mathbb{R}$, $\beta_t^{\hat{\pi}^*}: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$, and $\beta_t^*: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ such that

$$\beta_T^{\hat{\pi}^*}(h_T, a_T) \geq \Delta_T(h_T, a_T) \geq \beta_T^*(h_T, a_T) \quad (16)$$

and for $t < T$:

$$\beta_t^{\hat{\pi}^*}(h_t, a_t) \geq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (17)$$

$$\beta_t^*(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (18)$$

and for all $t \in \{1, \dots, T\}$:

$$\alpha_t^{\hat{\pi}^*}(h_t) \geq \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad (19)$$

$$\alpha_t^*(h_t) \leq \beta_t^*(h_t, \pi_t^*(h_t)), \quad (20)$$

where Δ_t is as defined in (12) (in particular, at terminal time, $\Delta_T(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T)$), and π_t^* and $\hat{\pi}_t^*$ are as in Lemma 2. Then, for all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$, the AIS-based policy $\pi_t(h_t) = \hat{\pi}_t^*(\sigma_t(h_t))$ satisfies

$$V_t^\pi(h_t) - V_t^*(h_t) \leq \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t). \quad (21)$$

Symmetrically, suppose that for each $t \in \{1, \dots, T\}$ there exist measurable functions $\underline{\alpha}_t^{\hat{\pi}^*}: \mathcal{H}_t \rightarrow \mathbb{R}$, $\underline{\alpha}_t^*: \mathcal{H}_t \rightarrow \mathbb{R}$, $\underline{\beta}_t^{\hat{\pi}^*}: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$, and $\underline{\beta}_t^*: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ such that

$$\underline{\beta}_T^{\hat{\pi}^*}(h_T, a_T) \leq \Delta_T(h_T, a_T) \leq \underline{\beta}_T^*(h_T, a_T) \quad (22)$$

and for $t < T$:

$$\underline{\beta}_t^{\hat{\pi}^*}(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (23)$$

$$\underline{\beta}_t^*(h_t, a_t) \geq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (24)$$

and for all $t \in \{1, \dots, T\}$:

$$\underline{\alpha}_t^{\hat{\pi}^*}(h_t) \leq \underline{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad (25)$$

$$\underline{\alpha}_t^*(h_t) \geq \underline{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \quad (26)$$

Then, for all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$,

$$V_t^\pi(h_t) - V_t^*(h_t) \geq \underline{\alpha}_t^{\hat{\pi}^*}(h_t) - \underline{\alpha}_t^*(h_t). \quad (27)$$

Proof. [Check proof]

Throughout the proof, let $\tilde{\alpha}_t^{\hat{\pi}^*}$, $\tilde{\beta}_t^{\hat{\pi}^*}$, $\tilde{\alpha}_t^*$, and $\tilde{\beta}_t^*$ denote the *exact* signed error functions of Lemmas 1 and 2:

$$\tilde{\alpha}_t^{\hat{\pi}^*}(h_t) = V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \quad \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t) = Q_t^\pi(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t),$$

i.e., the functions of Lemma 1 specialized to $\hat{\pi} = \hat{\pi}^*$, and

$$\tilde{\alpha}_t^*(h_t) = V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \quad \tilde{\beta}_t^*(h_t, a_t) = Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t),$$

i.e., the functions of Lemma 2. Regarding the specialization $\hat{\pi} = \hat{\pi}^*$ in Lemma 1: since $\hat{\pi}_t^*(z_t)$ is a minimizer of $\hat{Q}_t^*(z_t, \cdot)$, evaluating the AIS dynamic program along $\hat{\pi}^*$ shows that the policy-evaluation value functions of $\hat{\pi}^*$ coincide with the optimal ones, $\hat{V}_t^{\hat{\pi}^*} = \hat{V}_t^*$ and $\hat{Q}_t^{\hat{\pi}^*} = \hat{Q}_t^*$ for all t ; consequently, the integrand $\hat{V}_{t+1}^{\hat{\pi}^*}$ in (3) equals $\hat{V}_{t+1}^*$ and the mismatch terms coincide, $\Delta_t^{\hat{\pi}^*} = \Delta_t$, the same Δ_t as in Lemma 2 and in the statement of the theorem. Thus Lemmas 1 and 2 give, for all t , h_t , and a_t (with $\tilde{\alpha}_{T+1}^{\hat{\pi}^*} \equiv 0$ and $\tilde{\alpha}_{T+1}^* \equiv 0$):

$$\begin{aligned} \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \\ \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) &= \tilde{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \end{aligned} \quad (28)$$

$$\begin{aligned} \tilde{\beta}_t^*(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \\ \tilde{\beta}_t^*(h_t, \pi_t^*(h_t)) &\leq \tilde{\alpha}_t^*(h_t). \end{aligned} \quad (29)$$

Step 1 (the candidate functions compared to the exact errors). We show by backward induction that for all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$,

$$\alpha_t^{\hat{\pi}^*}(h_t) \geq \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) \quad \text{and} \quad \alpha_t^*(h_t) \leq \tilde{\alpha}_t^*(h_t). \quad (30)$$

Base case ($t = T$). Since $\tilde{\alpha}_{T+1}^{\hat{\pi}^*} \equiv 0$ and $\tilde{\alpha}_{T+1}^* \equiv 0$, the recursions in (28) and (29) give $\tilde{\beta}_T^{\hat{\pi}^*} = \tilde{\beta}_T^* = \Delta_T$. Hence, by (16), $\beta_T^{\hat{\pi}^*} \geq \tilde{\beta}_T^{\hat{\pi}^*}$ and $\beta_T^* \leq \tilde{\beta}_T^*$. Therefore, by (19) and the identity in (28),

$$\alpha_T^{\hat{\pi}^*}(h_T) \geq \beta_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \geq \tilde{\beta}_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) = \tilde{\alpha}_T^{\hat{\pi}^*}(h_T),$$

and, by (20) and the sandwich inequality in (29),

$$\alpha_T^*(h_T) \leq \beta_T^*(h_T, \pi_T^*(h_T)) \leq \tilde{\beta}_T^*(h_T, \pi_T^*(h_T)) \leq \tilde{\alpha}_T^*(h_T).$$

Induction step ($t < T$). Suppose (30) holds at $t + 1$. Then, by (17), the induction hypothesis, and the recursion in (28),

$$\begin{aligned} \beta_t^{\hat{\pi}^*}(h_t, a_t) &\geq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\geq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t), \end{aligned}$$

and then, exactly as in the base case, $\alpha_t^{\hat{\pi}^*}(h_t) \geq \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \geq \tilde{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t)$. Symmetrically, by (18), the induction hypothesis, and the recursion in (29),

$$\begin{aligned}\beta_t^*(h_t, a_t) &\leq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\leq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^*(h_t, a_t),\end{aligned}$$

and then $\alpha_t^*(h_t) \leq \beta_t^*(h_t, \pi_t^*(h_t)) \leq \tilde{\beta}_t^*(h_t, \pi_t^*(h_t)) \leq \tilde{\alpha}_t^*(h_t)$. This establishes (30) for all t .

Step 2 (bridge identity). Adding and subtracting $\hat{V}_t^*(\sigma_t(h_t))$,

$$V_t^\pi(h_t) - V_t^*(h_t) = [V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t))] - [V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t))] = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) - \tilde{\alpha}_t^*(h_t).$$

Combining Step 2 with (30),

$$V_t^\pi(h_t) - V_t^*(h_t) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) - \tilde{\alpha}_t^*(h_t) \leq \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t),$$

which is (21).

Step 3 (the lower bound). The argument mirrors Steps 1–2 with every inequality reversed; the only structural change is that the $\star$ -side now invokes the *upper* side of the sandwich in Lemma 2, namely

$$\tilde{\alpha}_t^*(h_t) \leq \tilde{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad (31)$$

in place of (29). We show by backward induction that for all t and all $h_t \in \mathcal{H}_t$,

$$\underline{\alpha}_t^{\hat{\pi}^*}(h_t) \leq \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) \quad \text{and} \quad \underline{\alpha}_t^*(h_t) \geq \tilde{\alpha}_t^*(h_t). \quad (32)$$

Base case ($t = T$). As before $\tilde{\beta}_T^{\hat{\pi}^*} = \tilde{\beta}_T^* = \Delta_T$, so by (22), $\underline{\beta}_T^{\hat{\pi}^*} \leq \tilde{\beta}_T^{\hat{\pi}^*}$ and $\underline{\beta}_T^* \geq \tilde{\beta}_T^*$. Therefore, by (25) and the identity in (28),

$$\underline{\alpha}_T^{\hat{\pi}^*}(h_T) \leq \underline{\beta}_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \leq \tilde{\beta}_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) = \tilde{\alpha}_T^{\hat{\pi}^*}(h_T),$$

and, by (26) and the upper sandwich inequality (31),

$$\underline{\alpha}_T^*(h_T) \geq \underline{\beta}_T^*(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \geq \tilde{\beta}_T^*(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \geq \tilde{\alpha}_T^*(h_T).$$

Induction step ($t < T$). Suppose (32) holds at $t + 1$. Then, by (23), the induction hypothesis, and the recursion in (28),

$$\begin{aligned}\underline{\beta}_t^{\hat{\pi}^*}(h_t, a_t) &\leq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\leq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t),\end{aligned}$$

and then, exactly as in the base case, $\underline{\alpha}_t^{\hat{\pi}^*}(h_t) \leq \underline{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \leq \tilde{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t)$. Symmetrically, by (24), the induction hypothesis, and the recursion in (29),

$$\begin{aligned}\beta_t^*(h_t, a_t) &\geq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\geq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^*(h_t, a_t),\end{aligned}$$

and then $\underline{\alpha}_t^*(h_t) \geq \underline{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \geq \tilde{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \geq \tilde{\alpha}_t^*(h_t)$, where the last inequality is the upper sandwich side (31). This establishes (32) for all t . Combining with the bridge identity of Step 2,

$$V_t^\pi(h_t) - V_t^*(h_t) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) - \tilde{\alpha}_t^*(h_t) \geq \underline{\alpha}_t^{\hat{\pi}^*}(h_t) - \underline{\alpha}_t^*(h_t),$$

which is (27). $\square$

Theorem 4. Let $\pi = \hat{\pi}^* \circ \sigma$ be the lifted AIS-optimal policy in the true POMDP. Define

$$\Delta_t(h_t, a_t) := \left[c_t(h_t, a_t) + \int \hat{V}_{t+1}^*(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t) \right] - \left[\hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^*(z_{t+1}) \hat{P}_t^\sigma(dz_{t+1} | h_t, a_t) \right],$$

and

$$\begin{aligned} \alpha_t^{\hat{\pi}^*}(h_t) &:= V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t)), & \alpha_t^*(h_t) &:= V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \\ \beta_t^{\hat{\pi}^*}(h_t, a_t) &:= Q_t^\pi(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t), & \beta_t^*(h_t, a_t) &:= Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t). \end{aligned}$$

For each $t \in \{1, \dots, T\}$ and each $h_t \in \mathcal{H}_t$, $a_t \in \mathcal{A}$, we have

Part 1 (Policy-evaluation)

$$\begin{aligned} \beta_t^{\hat{\pi}^*}(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \\ \alpha_t^{\hat{\pi}^*}(h_t) &= \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \end{aligned}$$

Part 2 (Optimal)

$$\begin{aligned} \beta_t^*(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \\ \beta_t^*(h_t, \pi_t^*(h_t)) &\leq \alpha_t^*(h_t) \leq \beta_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \end{aligned}$$

With the intermediate bounds above we can state the main result. For all $t \in \{1, \dots, T\}$ and $h_t \in \mathcal{H}$, the AIS-based policy $\pi = \hat{\pi}^*(\sigma_t(h_t))$ satisfies

$$V_t^\pi(h_t) - V_t^*(h_t) = \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t) \leq \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) - \beta_t^*(h_t, \pi_t^*(h_t)). \quad (33)$$

and

$$V_t^\pi(h_t) - V_t^*(h_t) = \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t) \geq \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) - \beta_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \quad (34)$$

Lower bound We can define $\underline{\alpha}_t^*(h_t)$ such that

$$\underline{\alpha}_t^*(h_t) \leq \beta_t^*(h_t, \pi_t^*(h_t)).$$

To do so, we need to identify a $\underline{\beta}_t^*(h_t, a_t) \leq \beta_t^*(h_t, a_t)$. Therefore, we consider the following recursion:

$$\underline{\beta}_t^*(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t),$$

and

$$\underline{\alpha}_t^*(h_t) \leq \underline{\beta}_t^*(h_t, \pi_t^*(h_t))$$

Upper bound By the same argument, the following is an upper bound:

$$\bar{\beta}_t^*(h_t, a_t) \geq \Delta_t(h_t, a_t) + \int \bar{\alpha}_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t),$$

and

$$\bar{\alpha}_t^*(h_t) \geq \bar{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t)))$$

Unified I want to claim that

$$|\alpha_t^*(h_t)| \leq A_t^*(h_t).$$

To do so, we first find an upper bound on β_t^* , i.e.,

$$|\beta_t^*(h_t, a_t)| \leq B_t^*(h_t, a_t)$$

I suspect that the following should work.

$$B_t^*(h_t, a_t) \geq |\Delta_t(h_t, a_t)| + \int A_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t),$$

and

$$A_t^*(h_t) \geq \max\{B_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), B_t^*(h_t, \pi^*(h_t))\}$$

We should be able to argue that (A^*, B^*) follows the upper bound recursion and $(-A^*, -B^*)$ follows the lower bound recursion. Moreover, we would need to argue that these are upper and lower bounds for β^* as well.

Proof

Base case ($t = T$)

We have

$$Q_T^\pi(h_T, a_T) = Q_T^*(h_T, a_T) = c_T(h_T, a_T), \quad \hat{Q}_T^*(\sigma_T(h_T), a_T) = \hat{c}_T(\sigma_T(h_T), a_T).$$

Then,

$$\beta_T^*(h_T, a_T) = \beta_T^*(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T) = \Delta_T(h_T, a_T).$$

Part 1 at $t = T$. Since $V_T^\pi(h_T) = Q_T^\pi(h_T, \hat{\pi}_T^*(\sigma_T(h_T)))$ and $\hat{V}_T^*(\sigma_T(h_T)) = \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T)))$, we have

$$\alpha_T^*(h_T) = Q_T^\pi(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) - \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T))) = \beta_T^*(h_T, \hat{\pi}_T^*(\sigma_T(h_T))).$$

Part 2 at $t = T$.

$$\beta_T^*(h_T, a_T) = Q_T^*(h_T, a_T) - \hat{Q}_T^*(\sigma_T(h_T), a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T) = \Delta_T(h_T, a_T)$$

The two value functions evaluate at different actions:

$$V_T^*(h_T) = Q_T^*(h_T, \pi_T^*(h_T)), \quad \hat{V}_T^*(\sigma_T(h_T)) = \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T)))$$

Upper bound

$$\begin{aligned} \alpha_T^*(h_T) &= V_T^*(h_T) - \hat{V}_T^*(\sigma_T(h_T)) \\ &= Q_T^*(h_T, \pi_T^*(h_T)) - \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T))) \\ &\leq Q_T^*(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) - \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T))) = \beta_T^*(h_T, \hat{\pi}_T^*(\sigma_T(h_T))). \end{aligned}$$

Lower bound

$$\begin{aligned} \alpha_T^*(h_T) &= V_T^*(h_T) - \hat{V}_T^*(\sigma_T(h_T)) \\ &= Q_T^*(h_T, \pi_T^*(h_T)) - \hat{Q}_T^*(\sigma_T(h_T), \hat{\pi}_T^*(\sigma_T(h_T))) \\ &\geq Q_T^*(h_T, \pi_T^*(h_T)) - \hat{Q}_T^*(\sigma_T(h_T), \pi_T^*(h_T)) = \beta_T^*(h_T, \pi_T^*(h_T)). \end{aligned}$$

Combining, we have

$$\beta_T^*(h_T, \pi_T^*(h_T)) \leq \alpha_T^*(h_T) \leq \beta_T^*(h_T, \hat{\pi}_T^*(\sigma_T(h_T))).$$

Induction step ($t < T$)

Inductive hypothesis. *At time $t + 1$:*

- $\alpha_{t+1}^{\hat{\pi}^*}(h_{t+1}) = \beta_{t+1}^{\hat{\pi}^*}(h_{t+1}, \hat{\pi}_{t+1}^*(\sigma_{t+1}(h_{t+1})))$
- $\beta_{t+1}^*(h_{t+1}, \pi_{t+1}^*(h_{t+1})) \leq \alpha_{t+1}^*(h_{t+1}) \leq \beta_{t+1}^*(h_{t+1}, \hat{\pi}_{t+1}^*(\sigma_{t+1}(h_{t+1})))$

Part 1: the $\beta^{\hat{\pi}^*}$ recursion. *By definition,*

$$\begin{aligned}
\beta_t^{\hat{\pi}^*}(h_t, a_t) &= Q_t^\pi(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t) \\
&= [c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)] \\
&\quad + \int V_{t+1}^\pi(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) - \int \hat{V}_{t+1}^*(z_{t+1}) \hat{P}_t(dz_{t+1} \mid \sigma_t(h_t), a_t) \\
&= [c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)] + \underbrace{\int \hat{V}_{t+1}^*(z_{t+1}) P_t^\sigma(z_{t+1} \mid h_t, a_t) - \int \hat{V}_{t+1}^*(z_{t+1}) \hat{P}_t(dz_{t+1} \mid \sigma_t(h_t), a_t)}_{\Delta_t(h_t, a_t)} \\
&\quad + \int V_{t+1}^\pi(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) - \int \hat{V}_{t+1}^*(z_{t+1}) P_t^\sigma(z_{t+1} \mid h_t, a_t) \\
&= \Delta_t(h_t, a_t) + \int V_{t+1}^\pi(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) - \int \hat{V}_{t+1}^*(\sigma_t(h_{t+1})) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\
&= \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t).
\end{aligned}$$

1 Reward shaping

Let $\bar{c}_t(h_t, a_t) = c_t(h_t, a_t) + b_t$. This means,

$$\bar{Q}_t^\pi(h_t, a_t) = Q_t^\pi(h_t, a_t) + \sum_{\tau=t}^T b_t, \quad \bar{V}_t^\pi(h_t) = V_t^\pi(h_t) + \sum_{\tau=t}^T b_t$$

Let

$$\begin{aligned} \bar{\alpha}_t^{\hat{\pi}}(h_t) &= \bar{V}_t^\pi(h_t) - \hat{V}_t^{\hat{\pi}}(\sigma_t(h_t)) \\ \bar{\beta}_t^{\hat{\pi}}(h_t, a_t) &= \bar{Q}_t^\pi(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t) \end{aligned}$$

Define

$$\begin{aligned} \bar{\Delta}_t^{\hat{\pi}}(h_t, a_t) &= \left[\bar{c}_t(h_t, a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t) \right] \\ &\quad - \left[\hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^\sigma(dz_{t+1} | h_t, a_t) \right] \\ &= b_t + \text{Tr}(\Sigma P_{t+1}) \\ &= 0 \quad (\text{assum. } b_t = -\text{Tr}(\Sigma P_{t+1})). \end{aligned}$$

From the lemma, we have $\bar{\beta}_T^{\hat{\pi}}(h_t, a_t) = b_T = 0$ and $\bar{\alpha}_T^{\hat{\pi}}(h_t) = b_T = 0$. Then,

$$\begin{aligned} \bar{\beta}_t^{\hat{\pi}}(h_t, a_t) &= \int \bar{\alpha}_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) = 0 \\ \bar{\alpha}_t^{\hat{\pi}}(h_t) &= 0 \end{aligned}$$

Consequently,

$$\hat{V}_t^{\hat{\pi}}(\sigma_t(h_t)) = \bar{V}_t^\pi(h_t) + \bar{\alpha}_t = \bar{V}_t^\pi(h_t) = V_t^\pi(h_t) + \sum_{\tau=t}^T b_t$$